

VIVA QUESTIONS AND ANSWERS

1. Define Network? ✓

A network is a set of devices connected by physical media links. A network is recursively is connection of two or more nodes by a physical link or two or more networks connected by one or more nodes.

2. What is a Link?

At the lowest level, a network can consist of two or more computers directly connected by some physical medium such as coaxial cable or optical fiber. Such a physical medium is called as Link.

3. What is node?

A network can consist of two or more computers directly connected by physical medium such as coaxial cable or optical fiber. Such a physical medium is called as Links and the computer connects is called as Nodes.

4. What is gateway or Router? ✓

A node that is connected to two or more networks is commonly called as router or Gateway. It generally forwards message from one network to another.

5. What is point-point link? ✓

If the physical links are limited to a pair of nodes it is said to be point to point link.

6. What is Multiple Accesses? ✓

If the physical links are shared by more than two nodes, it is said to be Multiple Access.

7. What are the criteria necessary for an effective and efficient network?

- a) Performance - It can be measured in many ways, including transmit time and response time.
- b) Reliability - It is measured by frequency of failure, the time it takes a link to recover from a failure, and the network's robustness.
- c) Security - Security issues include protecting data from unauthorized access and virus.

8. What is protocol? ✓

A protocol is a set of rules that govern all aspects of information communication.

9. What are the key elements of protocols? ✓

The key elements of protocols are

- a) Syntax - It refers to the structure or format of the data, that is the order in which they are presented.
- b) Semantics - It refers to the meaning of each section of bits.

c) Timing - Timing refers to two characteristics: when data should be sent and how fast they can be sent.

10. What is a peer-peer process? ✓

The processes on each machine that communicate at a given layer are called peer-peer process.

11. Define Bandwidth and Latency?

Network performance is measured in Bandwidth (throughput) and Latency (Delay). Bandwidth of a network is given by the number of bits that can be transmitted over the network in a certain period of time. Latency corresponds to how long it takes a message to travel from one end of a network to the other. It is strictly measured in terms of time.

12. What is Round Trip Time?

The duration of time takes to send a message from one end of a network to the other back, is called RTT.

13. What is IP address? ✓

An Internet Protocol address (IP address) is a numerical label assigned to each device (e.g., computer, printer) participating in a computer network that uses the Internet Protocol for communication. An IP address serves two principal functions: host or network interface identification and location addressing.

14. What are IP versions? ✓

Two versions of the Internet Protocol (IP) are in use: IP Version 4 and IP Version 6. Each version defines an IP address differently. Because of its prevalence, the generic term IP address typically still refers to the addresses defined by IPv4.

15. What is an IPv4 address? ✓

Decomposition of an IPv4 address from dot-decimal notation to its binary value. In IPv4 an address consists of 32 bits which limits the address space to 4294967296 (2^{32}) possible unique addresses. IPv4 reserves some addresses for special purposes such as private networks (~18 million addresses) or multicast addresses (~270 million addresses).

16. What is an IPv6 address? ✓

The rapid exhaustion of IPv4 address space prompted the Internet Engineering Task Force (IETF) to explore new technologies to expand the addressing capability in the Internet. The permanent solution was deemed to be a redesign of the Internet Protocol itself. This new generation of the Internet Protocol was eventually named Internet Protocol Version 6 (IPv6) in 1995. The address size was increased from 32 to 128 bits (16 octets), thus providing up to

2^{128} (approximately 3.403×10^{38}) addresses. This is deemed sufficient for the foreseeable future.

17. What are Methods in ip address? ✓

- Uses of dynamic address assignment
- Sticky dynamic IP address
- Address auto configuration
- Uses of static addressing
- Conflict
- Modifications to IP addressing?
- IP blocking and firewalls
- IP address translation

18. What is Socket? ✓

A socket is an endpoint between two way communication.

19. What is Socket class?

A socket is simply an endpoint for communications between the machines. The Socket class can be used to create a socket.

20. What is ServerSocket class? ✓

The ServerSocket class can be used to create a server socket. This object is used to establish communication with the clients.

21. What is Echo Command? ✓

In echo is a command in DOS, OS/2, Microsoft Windows, Singularity, Unix and Unix-like operating systems that places a string on the computer terminal. It is a built-in command typically used in shell scripts and batch files to output status text to the screen or a file. Many shells, including Bash and zsh, implement echo as a builtin command.

22. What is ping command? ✓

The ping command is a Command Prompt command used to test the ability of the source computer to reach a specified destination computer. The ping command is usually used as a simple way verify that a computer can communicate over the network with another computer or network device.

23. What are the Ping Related Commands?

The ping command is often used with other networking related Command Prompt commands like tracert, ipconfig, netstat, nslookup, and others.

24. Is there a continuous ping options?

ping <address> -t

Use the -t option to ping any address until you cancel it by pressing Ctrl +C.

25. IP address of ping?

The user would first start by using the ping command to ping the IP address 204.228.150.3.

26. What is meant by Talk command? ✓

Talk is a visual communication program which copies lines from your terminal to that of another user, much like an instant messenger service.

27. Define the term unicasting, Multicasting and Broadcasting?

- If the message is sent from a source to a single destination node, it is called unicasting.
- If the message is sent to some subset of other nodes, it is called Multicasting.
- If the message is sent to all the m nodes in the networking it is called Broadcasting.

28. List the layers of OSI? ✓

- a) Physical Layer
- b) Data Link Layer
- c) Network Layer
- d) Transport Layer
- e) Session Layer
- f) Presentation Layer
- g) Application Layer

29. Which layers are network support layers?

- a) Physical Layer
- b) Data Link Layer
- c) Network Layer

30. Which layers are user support layers?

- a) Session Layer
- b) Presentation Layer
- c) Application Layer

31. What are the responsibilities of Data Link Layer? ✓

The Data Link Layer transforms the physical layer, a raw transmission facility, to a reliable link and is responsible for node-node delivery.

- a) Framing

d) Directory Services

37. What are the types of errors?

a) Single Bit error

In a Single bit error only one bit in the data unit has changes.

b) Burst error

A Burst error means that two or more bits in the data have changed.

38. What is Redundancy?

The concept of including extra information in the transmission solely for the purpose of comparison. This technique is called redundancy.

39. What is CRC? ✓

CRC, is the most powerful of the redundancy checking techniques, is based on binary division.

40. Define CRC generator?

A CRC generator uses a modulo-2 division. In the first step, the 4-bit divisor is subtracted from the first 4 bit of the dividend. Each bit of the divisor is subtracted from the corresponding bit of the dividend with disturbing the next higher bit.

41. What is the implementation of CRC long division?

The CRC code is generated through a process of long division. This can be implemented using a digital logic circuit consisting of exclusive-or (XOR) gates and a shift register.

42. What is the use of CRC in USB?

The USB specification calls for the use of Cyclic Redundancy Checksums (CRC) to protect all non-PID fields in token and data packets from errors during transmission. This paper describes the mathematical basis behind CRC in an intuitive fashion and then explains the specific implementation called for in USB.

43. Write the application of CRC?

A CRC-enabled device calculates a short, fixed-length binary sequence, known as the check value or CRC, for each block of data to be sent or stored and appends it to the data, forming codeword.

44. What is a Polynomial Representation of Binary Words ?

For a coder works, let us first define the notion of the polynomial

To define a code we have to state how to obtain the polynomial $R(X)$ (that is, the appended bits) corresponding to any set of message bits or $M(X)$. The appended bits are called the frame check sequence for the data frame of k bits.

46. Write few Importance of Error Detection? ✓

A transmitted bit can be received in error, due to "noise" on the transmission channel. If we are dealing with voice or video data, the occurrence of errors in a small percentage of bits is quite tolerable, but in many other cases it is crucial that all bits be received intact.

47. Give the essential properties for polynomial?

A polynomial should be selected to have at least the following properties.

a) It should not be

b) It should be divisible by $(x+1)$.

48. What is Checksum? ✓

Checksum is used by the higher layer protocols (TCP/IP) for error detection.

49. Define Retransmission?

Retransmission is a technique in which the receiver detects the occurrence of an error and asks the sender to resent the message. Resending is repeated until a message arrives that the receiver believes is error freed.

50. What is framing? ✓

Framing in the data link layer separates a message from one source to a destination, or from other messages to other destinations, by adding a sender address and a destination address. The destination address defines where the packet has to go and the sender address helps the recipient acknowledge the receipt

51. What is Flow control?

Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgement.

52. What is stop and wait protocol? ✓

In Stop and wait protocol, sender sends one frame, wait until it receives confirmation from the receiver (okay to go ahead), and then sends the next frame.

53. What is sliding window? ✓

The sliding window is an abstract concept that defines the range of sequence numbers that is the concern of the sender and receiver. In other words, the sender and receiver need to deal with only part of the possible sequence numbers.

54. What is Piggy Backing? ✓

...ve repeat. ...ing errors when frames are pipelined is called

62. How many buffers must the receiver have?

Under no conditions will it ever accept frames whose sequence numbers are below the lower edge of the window or frames whose sequence numbers are above the upper edge of the window. Consequently, the number of buffers needed is equal to the window size, not to the range of sequence numbers.

63. What is meant by acknowledgement?

An acknowledgement is a signal passed between communicating Processes or computer To signify acknowledgement, or receipt of response, as part of a communications protocol.

64. What is TCP socket?

In TCP/IP and UDP networks, a port is an endpoint to a logical connection and the way a client program specifies a specific server program on a computer in a network. This list of well-known port numbers specifies the port used by the server process as its contact port.

65. What is UDP socket?

UDP is a connectionless protocol and the socket is created for client and server to transfer the data. Socket connection is achieved using the port number

66. What is DNS? ✓

DNS stands for domain name system. Unique name of the host is identified with its IP address through server client communication.

67. What is raw socket?

Raw socket is similar to the TCP or UDP socket and raw socket is used to capture the packet of data and filter packet with error.

68. What is data frame?

A data frame is an aggregate of numerous, partly overlapping collections of data and metadata that have been derived from massive amounts of network activity such as Content production, consumption, and other user behavior.

69. What is subnet?

A generic term for section of a large networks usually separated by a bridge or router.

70. What are the possible ways of data exchange?

- Simplex
- Half-duplex
- Full-duplex

The address for a device as it is identified at the Media Access Control (MAC) layer in the network architecture. MAC address is usually stored in ROM on the network adapter card and is unique.

93. What is difference between ARP and RARP?

The address resolution protocol is used to associate the 32 bit IP address with the 48 bit physical address, used by a host or a router to find the physical address of another host on its network by sending a ARP query packet that includes the IP address of the receiver.

The reverse address resolution protocol allows a host to discover its Internet address when it knows only its physical address.

94. What is multicast routing?

Sending a message to a group is called multicasting, and its routing algorithm is called multicast routing.

95. What are the important topologies for networks? ✓

- Bus topology: In this each computer is directly connected to primary network cable in a single line. Advantage of bus topology is an Inexpensive and easy to install.
- Star topology: In this all computers are connected using a central hub. Advantage of star topology is easy to install and also easy to trouble shoot physical problems.
- Ring topology: In this all computers are connected in loop. Advantage of ring topology is all the computers have equal access to network media.

96. What is the range of address in the classes of internet address?

- Class A – 0.0.0.0 to 127.255.255.255
- Class B – 128.0.0.0 to 191.255.255.255
- Class C – 192.0.0.0 to 223.255.255.255
- Class D – 224.0.0.0 to 239.255.255.255
- Class E – 240.0.0.0 to 247.255.255.255

97. What is called multipoint communication?

Some applications require data to be delivered from a sender to multiple receivers. Examples of such applications include audio and video broadcasts, real-time delivery of stock quotes, and teleconferencing applications. A service where data is delivered from a sender to multiple receivers is called multipoint communication or multicast.